

COMMONWEALTH OF AUSTRALIA

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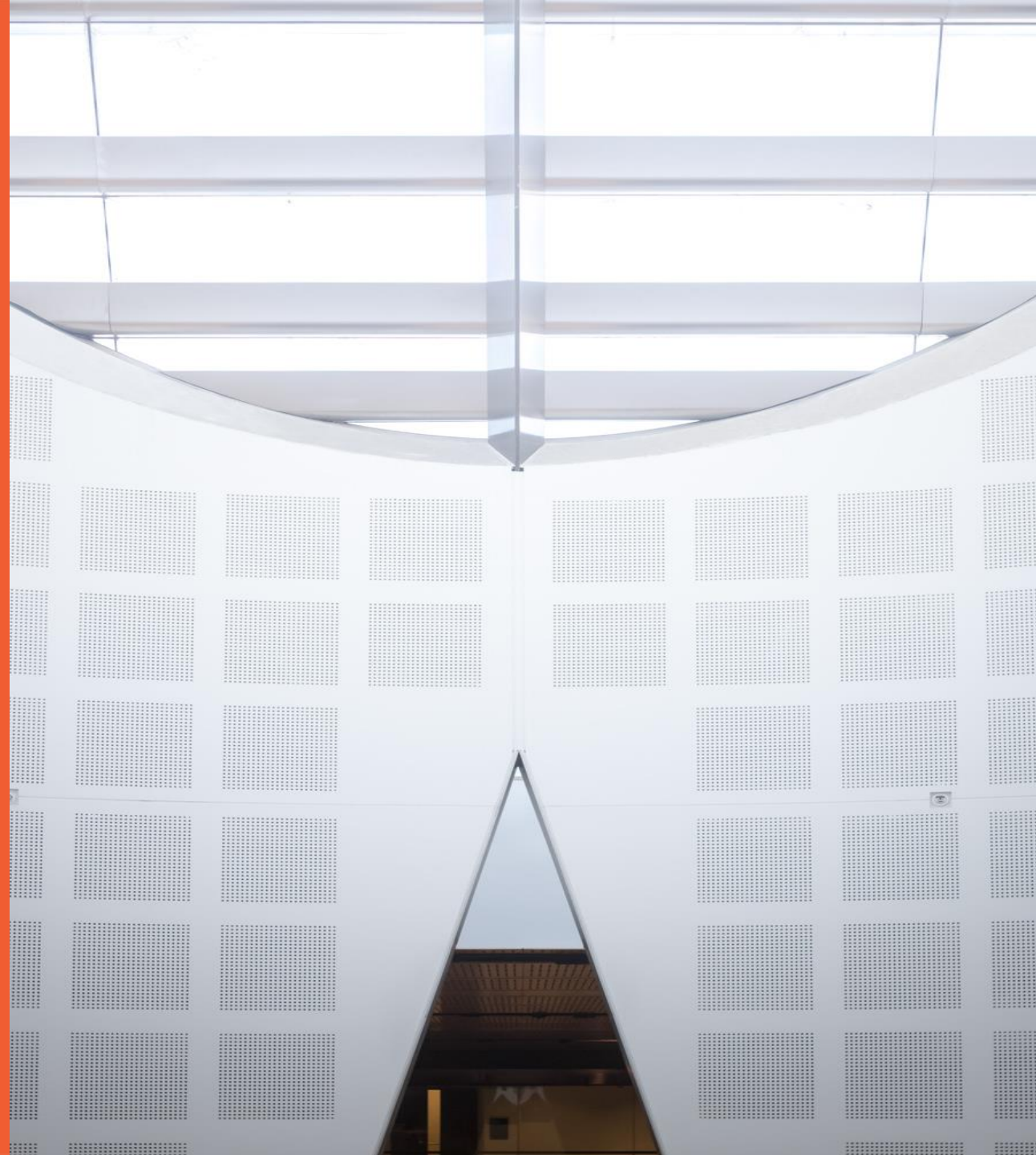
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COMPx270: Randomised and
Advanced Algorithms
Lecture 12: Learning from
experts

Clément Canonne
School of Computer Science



THE UNIVERSITY OF
SYDNEY



A question

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However, you don't know anything about the stock market.

But you have many friends who do: they're all "experts."

A question

So every morning, before you make your decision, all those friends will give you their advice.



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So every morning, before you make your decision, all those friends will give you their advice.



Some might collude, or be completely wrong, or even try to make you lose money. But each of them will tell you to either **sell** or **buy**.

A question

Then, based on those many pieces of advice, **you** decide.

(And you do that again, every day.)



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What is a good strategy to make money?



A question

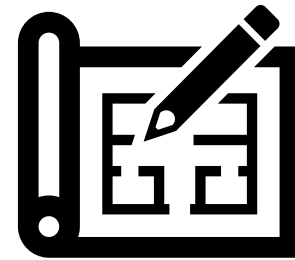
Then, based on those many pieces of advice, **you** decide.

(And you do that again, every day.)

What is a **provably** good strategy to make money?



Let's make this formal



- There are n experts.
- Each day, $t = 1, \dots, T$, each of them makes a prediction $v_{i,t} \in \{0,1\}$
- Based on those, you make your own prediction $\hat{u}_t \in \{0,1\}$
- Then the “true” value $u_t \in \{0,1\}$ is revealed
- If $\hat{u}_t \neq u_t$, this counts as a **mistake** (mistakes are bad)

possibly infinite

expert:

day t



Goal: minimise **total number of mistakes** $M = \sum_{t=1}^T 1_{\hat{u}_t \neq u_t}$

- There are n experts.
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Goal: minimise **total number of mistakes** $M = \sum_{t=1}^T 1_{\hat{u}_t \neq u_t}$

But what do we mean by this? We don't assume **anything** on the experts or on the true values. **They could even all be adversarial!**

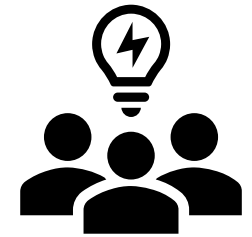
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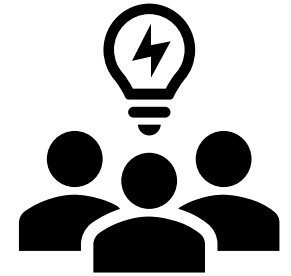
Goal: minimise **total number of mistakes** $M = \sum_{t=1}^T 1_{\hat{u}_t \neq u_t}$ compared to the **best expert** (whoever that is).

Not make much more mistakes than the **best advice in hindsight**.

Warmup: a Perfect Expert

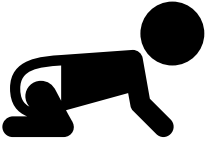


- There are n experts. **Suppose one of them (unknown) is always right.**
- Each day, $t = 1, \dots, T$, each of them makes a prediction $v_{i,t} \in \{0,1\}$
- Based on those, you make your own prediction $\hat{u}_t \in \{0,1\}$
- Then the “true” value $u_t \in \{0,1\}$ is revealed
- If $\hat{u}_t \neq u_t$, this counts as a **mistake** (mistakes are bad)



Goal: minimise **total number of mistakes** $M = \sum_{t=1}^T 1_{\hat{u}_t \neq u_t}$

Theorem. There is a strategy guaranteeing $M \leq n - 1$, regardless of T (even for $T = \infty$).



S: pool of experts I still listen to

```
Set  $S \leftarrow [n]$ 
for all  $1 \leq t \leq T$  do
  Receive  $v_{1,t}, \dots, v_{n,t}$ 
  if  $|S| \geq 1$  then
    Pick any  $i \in S$                                 ▷ Lexicographically, for instance
    Choose  $\hat{u}_t \leftarrow v_{i,t}$ 
  else
    Choose  $\hat{u}_t \leftarrow 0$                                 ▷ Arbitrary
  Receive  $u_t$                                 ▷ Observe the truth
   $S \leftarrow S \setminus \{i \in S : v_{i,t} \neq u_t\}$     ▷ Remove all mistaken experts
```

remove from the pool every expert who was wrong

Theorem. There is a strategy guaranteeing $M \leq n - 1$, regardless of T (even for $T = \infty$).

Proof.

① Every time I make a mistake, $|S|$ decreases by at least 1.

② At first, $|S| = n$

③ Also, $|S| \geq 1$ always (there is a "perfect expert")

Potential argument:

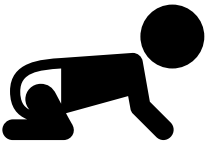
Potential function:

$$\phi = |S|$$

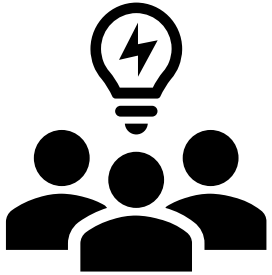
$$\phi_0 = n$$

$$\phi_t \geq 1 \quad \forall t$$

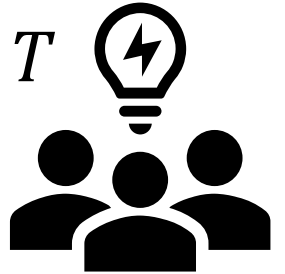
when mistake,
 $\phi_{t+1} \leq \phi_t - 1$



Theorem. There is a strategy guaranteeing $M \leq \log_2 n$, regardless of T (even for $T = \infty$).

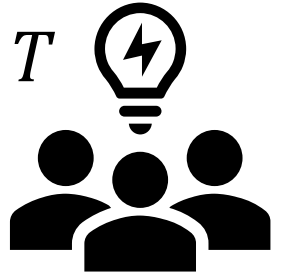


Claim. There is a strategy guaranteeing $M \leq \log_2 n$, regardless of T (even for $T = \infty$).



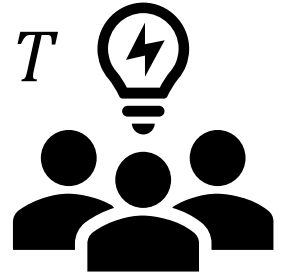
Algorithm: Start with $S = \{1, 2, \dots, n\}$. Each day, choose $\hat{u}_t$ to be the **majority** of advices from experts still in S . At the end of the day, remove from S all experts who predicted wrong.

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```
Set  $S \leftarrow [n]$ 
for all  $1 \leq t \leq T$  do
  Receive  $v_{1,t}, \dots, v_{n,t}$ 
  if  $|S| \geq 1$  then
    Choose  $\hat{u}_t \leftarrow \text{maj}_{i \in S} v_{i,t}$            ▷ Take the majority advice
  else
    Choose  $\hat{u}_t \leftarrow 0$                              ▷ Arbitrary
  Receive  $u_t$                                            ▷ Observe the truth
   $S \leftarrow S \setminus \{i \in S : v_{i,t} \neq u_t\}$    ▷ Remove all mistaken experts
```

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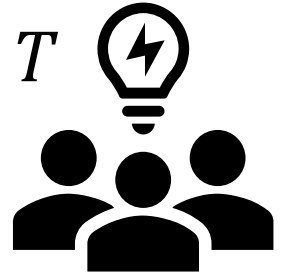
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Proof of correctness. Every time we make a mistake, at least half the experts in S must have been wrong (we took the majority vote). So after each mistake the size of S is at least halved.

$$\begin{aligned}\phi &= |S| \\ \phi_0 &= n \\ \phi_t &\geq 1 \quad \forall t\end{aligned}$$

$$\text{when mistake, } \phi_{t+1} \leq \frac{1}{2}\phi_t$$

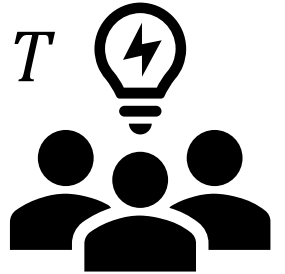
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Proof of correctness. Every time we make a mistake, **at least half** the experts in S must have been wrong (we took the majority vote). So after each mistake the size of S is at least **halved**. But we always have $|S| \geq 1$, since (by assumption) there exists an expert who is always right (and therefore never gets removed).

Claim. There is a strategy guaranteeing $M \leq \log_2 n$, regardless of T (even for $T = \infty$).



Algorithm: Start with $S = \{1, 2, \dots, n\}$. Each day, choose $\hat{u}_t$ to be the **majority** of advices from experts still in S . At the end of the day, remove from S all experts who predicted wrong.

Proof of correctness. Since we started with $|S| = n$, our total number M of mistakes must then satisfy

$$\frac{n}{2^M} \geq 1$$

HALVING ALGORITHM

that is, $M \leq \log_2 n$.

Nobody's Perfect



This is great! But... things completely fail if there is no “perfect expert.”

What if even the **best** expert made some mistakes? Can we make things **robust**?

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What if even the **best** expert made some mistakes? Can we make things **robust**?

Let's revisit the algorithm.

We had n **weights** $w_1, \dots, w_n$ initialised to 1.

At day t , our prediction was $\hat{u}_t \leftarrow \text{Maj}(w_1 v_{1,t} + \dots + w_n v_{n,t})$

Whenever expert i made a mistake, we set $w_i \leftarrow 0 \cdot w_i$.

$$S = \{i \in [n] : w_i = 1\}$$

weight 1: "Expert is in S "
weight 0: "Expert is not in S "

majority of predictions
over experts still in
 S

This is great! But... things completely fail if there is no “perfect expert.”

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Whenever expert i made a mistake, we set $w_i \leftarrow \frac{1}{2} \cdot w_i$.

Algorithm (Multiplicative Weights Update).

Start with n **weights** $w_1, \dots, w_n$ initialised to 1.

Each day, choose the **weighted majority** $\hat{u}_t \leftarrow \text{Maj}(w_1 v_{1,t} + \dots + w_n v_{n,t})$

At the end of the day, set $w_i \leftarrow \frac{1}{2} \cdot w_i$ for expert i made a mistake.

Set $w_1, \dots, w_n \leftarrow 1$
for all $1 \leq t \leq T$ do
 Receive $v_{1,t}, \dots, v_{n,t}$
 Choose $\hat{u}_t \leftarrow \text{sign}\left(\sum_{i=1}^n w_i v_{i,t} \geq \frac{1}{2} \sum_{i=1}^n w_i\right)$ $\triangleright$ Weighted majority
 Receive u_t $\triangleright$ Observe the truth
 for all $1 \leq i \leq n$ do $\triangleright$ Penalise all mistaken experts
 $w_i \leftarrow \begin{cases} \frac{1}{2} w_i & \text{if } v_{i,t} \neq u_t \\ w_i & \text{otherwise.} \end{cases}$

note of expert i
total current weight
current weight of expert i

$\text{sign}(x) = \begin{cases} 1 & \text{if } x \text{ true} \\ 0 & \text{o/w} \end{cases}$

```

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```

Theorem 60. *There is a (deterministic) algorithm (Algorithm 26) such that*

$$C(T) \leq \frac{C^*(T) + \log_2 n}{\log_2 \frac{4}{3}} \leq 2.41(C^*(T) + \log_2 n).$$

Moreover, this holds even when $T = \infty$.

errors of the best expert in hindsight
 $C^*(T) = \min_{1 \leq i \leq n} \sum_{t=1}^T \mathbb{1}_{v_{i,t} \neq u_t}$



Theorem. The MWU algorithm guarantees $M \leq 2.41(M^* + \log_2 n)$, where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

potential function

Proof. Let W_t be the total weights of experts on day t . Initially, $W_0 = n$. Every time we make a mistake, this means **at least half the weight** was on experts who did a mistake (since we took the weighted majority). So if we made a mistake at day t ,

$$W_{t+1} = W_t^{\text{good}} + \frac{1}{2} W_t^{\text{bad}} \leq \frac{1}{2} W_t + \frac{1}{2} \cdot \frac{1}{2} W_t = \frac{3}{4} W_t$$

weight of correct experts
weight of incorrect experts

Now, look at the **best expert** (in hindsight). They made M^* mistakes, so their final weight is $(1/2)^{M^*}$.

⑤ Let p be the fraction of weight on "good" experts:
 $W_t^{\text{good}} = p W_t$. Then ① $p \leq 1/2$ (I made a mistake!)

② $W_t = W_t^{\text{good}} + W_t^{\text{bad}}$

③ $W_t^{\text{good}} + \frac{1}{2} W_t^{\text{bad}} = p W_t + \frac{1}{2} (1-p) W_t = \frac{p+1}{2} W_t \leq \frac{1/2+1}{2} W_t = \frac{3}{4} W_t$

Theorem. The MWU algorithm guarantees $M \leq 2.41(M^* + \log_2 n)$, where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

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Theorem. The MWU algorithm guarantees $M \leq 2.41(M^* + \log_2 n)$, where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

$$\phi_0 = n$$

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$$\phi_{t+1} \leq \frac{3}{4} \phi_t \quad \text{when mistake}$$

$$W_{t+1} = W_t^{\text{good}} + \frac{1}{2} W_t^{\text{bad}} \leq \frac{1}{2} W_t + \frac{1}{2} \cdot \frac{1}{2} W_t = \frac{3}{4} W_t$$

Now, look at the **best expert** (in hindsight). They made M^* mistakes, so their final weight is $(1/2)^{M^*}$. So the final **total** weight is $W_T \geq (1/2)^{M^*}$.

$$\phi_t \geq \frac{1}{2^{M^*}} \quad \forall t$$

Theorem. The MWU algorithm guarantees $M \leq 2.41(M^* + \log_2 n)$, where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

Proof. Putting it all together:

$$\left(\frac{1}{2}\right)^{M^*} \leq W_T \leq \left(\frac{3}{4}\right)^{\overset{\text{\# of mistakes I make}}{M}} W_0 = \left(\frac{3}{4}\right)^M n$$

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Now, we take the logarithm:

$$-M^* \leq \underbrace{M \log_2 \left(\frac{3}{4}\right)}_{\leq 0} + \log_2 n$$

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$$-M^* \leq M \log_2 \left(\frac{3}{4}\right) + \log_2 n$$

and get

$$M \leq \frac{M^* + \log_2 n}{-\log_2 \left(\frac{3}{4}\right)}$$

Theorem. The MWU algorithm guarantees $M \leq 2.41(M^* + \log_2 n)$, where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

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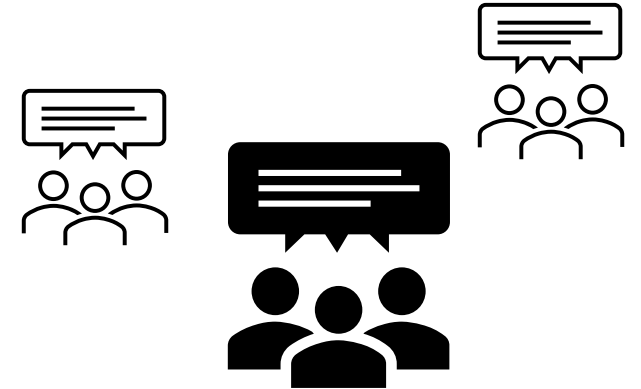
$$-M^* \leq M \log_2 \left(\frac{3}{4}\right) + \log_2 n$$

and get

$$M \leq \frac{M^* + \log_2 n}{-\log_2 \left(\frac{3}{4}\right)} \leq 2.41(M^* + \log_2 n)$$



Let's go further!



This is what we proved:

Algorithm (Multiplicative Weights Update).

Start with n **weights** $w_1, \dots, w_n$ initialised to 1.

Each day, choose the **weighted majority** $\hat{u}_t \leftarrow \text{Maj}(w_1 v_{1,t} + \dots + w_n v_{n,t})$

At the end of the day, set $w_i \leftarrow \frac{1}{2} \cdot w_i$ for expert i made a mistake.

Theorem. The MWU algorithm guarantees $M \leq 2.41(M^* + \log_2 n)$, where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

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Theorem. The MWU algorithm guarantees $M \leq 2.41(M^* + \log_2 n)$, where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

Using exactly the same argument (*try it!*), we get, for any $\beta \in (0,1)$:

Algorithm (Multiplicative Weights Update).

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Each day, choose the **weighted majority** $\hat{u}_t \leftarrow \text{Maj}(w_1 v_{1,t} + \dots + w_n v_{n,t})$

At the end of the day, set $w_i \leftarrow \beta \cdot w_i$ for expert i made a mistake.

Theorem. The MWU algorithm guarantees $M \leq \frac{M^* \log_2(1/\beta) + \log_2 n}{\log_2(\frac{2}{1+\beta})}$,

where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

Using exactly the same argument we get, for any $\beta \in (0,1)$:

Theorem. The MWU algorithm guarantees

$$M \leq \frac{M^* \log_2 \left(\frac{1}{\beta} \right) + \log_2 n}{\log_2 \left(\frac{2}{1+\beta} \right)}$$

$\beta \rightarrow 0 \nearrow \sim M^* \log_2 \frac{1}{\beta} + \log_2 n$
 $\beta = \frac{1}{2} \rightarrow \frac{M^* + \log_2 n}{\log_2 \frac{4}{3}}$

where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

$\beta = 1 - \varepsilon \searrow \frac{M^* \ln \frac{1}{1-\varepsilon} + \ln n}{\ln \frac{2}{2-\varepsilon}}$
 $\varepsilon \rightarrow 0 \searrow \simeq \frac{M^* \varepsilon + \ln n}{\varepsilon/2} = 2M^* + O\left(\frac{\log n}{\varepsilon}\right)$

Using exactly the same argument we get, for any $\beta = 1 - \varepsilon \in (0,1)$:

Theorem. The MWU algorithm guarantees

$$M \leq \frac{M^* \log_2 \left(\frac{1}{\beta} \right) + \log_2 n}{\log_2 \left(\frac{2}{1 + \beta} \right)} \approx 2 \left(M^* + \frac{\ln n}{\varepsilon} \right)$$

where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

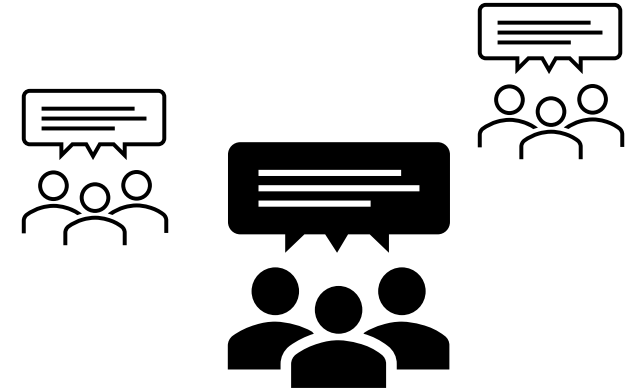
① $W_0 = n$

② $W_T \geq \beta^{M^*}$
↑ final weight of best expert

③ $W_{t+1} \leq W_t^{\text{good}} + \beta W_t^{\text{bad}} \leq \frac{1+\beta}{2} W_t$ when a mistake is made

$$\beta^{M^*} \leq W_T \leq \left(\frac{1+\beta}{2} \right)^M n$$

Is that tight?



Theorem. The MWU algorithm guarantees

$$M \approx 2 \left(M^* + \frac{\ln n}{\varepsilon} \right)$$

where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

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Can we improve that factor **2**?

Theorem. The MWU algorithm guarantees

$$M \approx 2 \left(M^* + \frac{\ln n}{\varepsilon} \right)$$

where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

Can we improve that factor **2**? **No.**

Theorem. The MWU algorithm guarantees

$$M \approx 2 \left(M^* + \frac{\ln n}{\varepsilon} \right)$$

where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

Can we improve that factor **2**? **No**. Consider two sets of $n/2$ experts, where experts in the first set are wrong on odd-numbered days, and those in the second set are wrong on even days. That will force T mistakes (while the best experts make $T/2$).

Theorem. The MWU algorithm guarantees

$$M \approx 2 \left(M^* + \frac{\ln n}{\varepsilon} \right)$$

where M^* is the # of mistakes made by the best expert. This holds regardless of T (even for $T = \infty$).

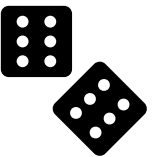
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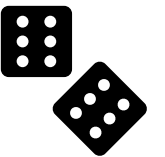


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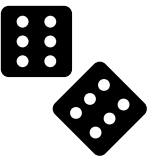


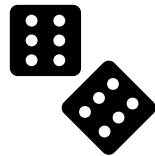
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Input: Penalty parameter $\beta \in (0, 1)$

Set $w_1, \dots, w_n \leftarrow 1$

for all $1 \leq t \leq T$ **do**

Receive $v_{1,t}, \dots, v_{n,t}$

Draw $I \in [n]$ according to the weights:

$$\Pr[I = i] = \frac{w_i}{\sum_{i=1}^n w_i}, \quad i \in [n]$$

Choose $\hat{u}_t \leftarrow v_{I,t}$

▷ One expert gets the vote

Receive u_t

▷ Observe the truth

for all $1 \leq i \leq n$ **do**

▷ Penalise all mistaken experts

$$w_i \leftarrow \begin{cases} \beta w_i & \text{if } v_{i,t} \neq u_t \\ w_i & \text{otherwise.} \end{cases}$$

key
change

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Theorem 62. *There is a (randomised) algorithm (Algorithm 28) such that*

$$\mathbb{E}[C(T)] \leq \frac{C^*(T) \ln(1/\beta) + \ln n}{1 - \beta}.$$

Moreover, this holds even when $T = \infty$.

Before:

$$C(T) \leq \frac{C^*(T) \ln(1/\beta) + \ln n}{\ln\left(\frac{2}{1+\beta}\right)}$$

Claim: $1 - \beta \geq \ln \frac{2}{1+\beta}$, so new bound is better!

Theorem 62. There is a (randomised) algorithm (Algorithm 28) such that

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Moreover, this holds even when $T = \infty$.

Pf. $\forall t, \mathbb{E}[C(t)] = \sum_{s=1}^t \Pr[\hat{u}_s \neq u_s] = \sum_{s=1}^t \frac{w_s^{\text{bad}}}{w_s} \triangleq F_s,$

the fraction of the weight on bad advice

At every time step,

$$W_t = \underbrace{(1 - F_t) W_t}_{\text{good weight}} + \underbrace{F_t W_t}_{\text{bad weight}}$$

$$\textcircled{3} W_{t+1} = (1 - F_t) W_t + \beta F_t W_t = W_t \cdot (1 - (1 - \beta) F_t)$$

$$\textcircled{1} W_0 = n$$

$$\textcircled{2} W_T \geq \beta^{C^*}$$

$$\beta^{C^*} \leq W_T \leq n \cdot \prod_{t=1}^T (1 - (1 - \beta) F_t)$$

"Take logarithms: $\prod \rightarrow \sum$ "

Recall $\mathbb{E}[C(T)] = \sum_{t=1}^T F_t$

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$$\begin{aligned} \frac{\ln(\beta^{C^*})}{-C^* \ln \frac{1}{\beta}} &\leq \ln n + \sum_{t=1}^T \ln(1 - (1-\beta)F_t) \\ C^* \ln \frac{1}{\beta} &\geq -\ln n + \sum_{t=1}^T -\ln(1 - (1-\beta)F_t) \\ &\geq -\ln n + (1-\beta) \underbrace{\sum_{t=1}^T F_t}_{\mathbb{E}[C(T)]} \end{aligned}$$

$$\text{so } \mathbb{E}[C(T)] \leq \frac{C^* \ln \frac{1}{\beta} + \ln n}{1 - \beta}$$

□

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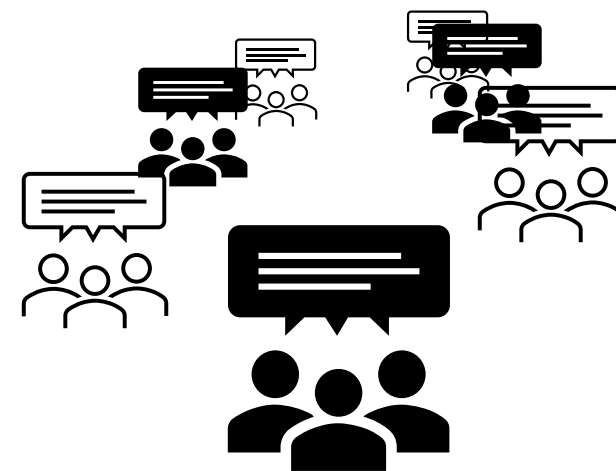
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Concluding remarks



- This was a **short** intro to the Multiplicative Weights Update Algorithms. Much more to say!
 - Different **predictions** (not only binary)
 - Different **payoffs** (not just 0-1 loss: correct/incorrect)
 - **Randomised** version!
- Discovered/rediscovered in many areas: **learning theory**, **game theory**/economics, **computational geometry**, **convex optimisation**...
- Many (sometimes unexpected) **applications**: online learning/bandits, semidefinite programming, flow algorithms, zero-sum games, algorithmic takes on evolution (!)

Some pointers if you have questions or want to know more about any of those (or connections to some of those topics):

- *The Multiplicative Weights Update Method: a Meta-Algorithm and Applications*. Arora, Hazan, Kale (2012):
<https://theoryofcomputing.org/articles/v008a006/>
- Lecture notes by Daniel Hsu (2017), Chapter 1:
<https://www.cs.columbia.edu/~djhsu/coms6998-f17/notes.pdf>

